

# Hazem Abuelanin

## Software Engineer

✉ hazem.abuelanin@gmail.com ☎ +201121909182 📍 Cairo, Egypt 🔗 Portfolio

🌐 LinkedIn profile 🛡 Exempted

## SUMMARY

---

**Software Engineer** specializing in AI and autonomous systems, with experience designing, integrating, and deploying perception, navigation, and embedded software across simulation and real-world platforms. Founded and led autonomous systems teams, delivering internationally competitive systems through end-to-end development spanning perception, planning, controls, and system integration.

## EXPERIENCE

---

### Robotics Software Engineer

10/2025 – 06/2026

*OmniLink*

Germany, Remotely

- Integrated OmniLink's AI platform onto real robot hardware and simulators, reducing manual validation by 40% through automated testing and benchmarking distributed system performance for early-stage feature development.

### Formula AI Engineer

09/2025 – Present

*ASU Racing Team*

Ain Shams University

- Designed and deployed a ROS 2 perception stack on NVIDIA Jetson AGX Xavier, implementing Visual-Inertial SLAM with the ZED2i, achieving 89% cone mapping accuracy, and processing real-time 32-channel Velodyne LiDAR data for autonomous perception.

### Research Assistant

08/2025 – Present

*Autotronics Research Lab (ARL)*

Ain Shams University

- Developed a low-level ESP32 motor control loop with micro-ROS integration for the F1TENTH platform and contributed to autonomous vehicle parameter estimation research through IPG CarMaker simulations, system identification, and literature-driven evaluation of estimation methods.

### Founder & Team Lead

2024 – 2025

*Shoubra Autonomous Racing Team*

- Founded and led a 20+ member engineering team to build the university's first autonomous racing team, overseeing architecture, LiDAR, computer vision, motion planning, and end-to-end system integration.

### Embedded systems engineer

2023 – 2024

*Shoubra racing team*

- Designed and validated a custom rear-distance alarm ECU using AVR microcontrollers, operating at >929% uptime during the 2024 Electric Vehicle Rally.

## EDUCATION

---

### Bachelor of Computer Engineering

09/2021 – 06/2026

*Benha university, Shoubra faculty of Engineering*

Cairo

GPA: 3.41/4.0

## SKILLS

---

### Programming

C - C++ - Python

### Robotics

ROS · ROS2 · Perception · 3D Sensing · LiDAR Processing · Point Clouds · Sensor Fusion · State Estimation · Control Systems · Sensor Calibration

### Embedded software engineering

Embedded C - Microcontrollers interfacing - Communication protocols - Microcontrollers architecture

### Artificial intelligence engineering

Data analysis - Machine learning - Deep learning - Computer vision

### Tools & Frameworks

CARLA - CoppeliaSim - Gazebo - Docker - Git - Linux - TensorFlow - PyTorch - Hugging Face

## PROJECTS

---

### Agri Inspection AMR (Si-Ware Sponsored Graduation Project)

02/2026 – 06/2026

- Built an autonomous agricultural inspection robot integrating a mobile base and robotic arm using ROS 2, Nav2, RP LiDAR A1, Intel RealSense D435, and MyCobot Pro 450, deployed on Jetson Orin Nano, Raspberry Pi, and ESP32.

### Fully autonomous vehicle for Shell Eco-marathon Competition

12/2024 – 05/2025

- Built an autonomous driving stack with computer vision, 3D LiDAR, sensor fusion, and motion planning in CARLA, validated through structured testing.

### Mobile Robot for Emirates Robotics Competition

12/2024 – 04/2025

- Built an autonomous navigation system using depth-camera point clouds, YOLOv10, sensor fusion, and AI-based path planning, with replay-driven debugging for robustness.

### Autonomous racing vehicle for F1TENTH competition

09/2024 – 10/2024

- Developed an autonomous racing stack using C++, ROS2, and LiDAR processing for real-time, reliable control in the AutoDRIVE simulator

### Machine Translation Model using Transformers

08/2024 – 10/2024

- Developed a neural translation model combining GRU + Transformer architectures with attention using TensorFlow/Python.

### Autonomous Vehicle for Electrical Vehicle Rally in Egypt

02/2024 – 07/2024

- Developed a ROS-based autonomous system in CoppeliaSim using 3D Velodyne LiDAR, monocular-depth cameras, and IMU, validated on a real autonomous vehicle.

### Distance alarm ECU

03/2024 – 04/2024

- Developed an ATmega32-based ECU for rear obstacle detection, implementing embedded firmware and validating operation on a competition vehicle.

## AWARDS

---

### Second place worldwide Shell Eco-marathon APC

2025

- Led Benha University's first Autonomous Racing Team to 2nd place in the Autonomous Urban Challenge among 24 international teams.

### Fourth Place worldwide F1TENTH IROS Abu Dhabi

2024

- Won 4th place in the autonomous racing challenge out of 58 international teams.

### Third place nationally Electric Vehicle Rally in Egypt

2024

- Won 3rd place in the autonomous driving track out of 15 Egyptian universities.

## TRAINING

---

### Microsoft Machine Learning Course

04/2024 – 10/2024

Digital Egypt Pioneers Initiative